

GuardianAI Engineering Handbook

Chapter 1 — Vision

Version: 1.0

Status: Draft for Review

Purpose

GuardianAI is an educational AI vision platform designed to help students understand how intelligent systems perceive, understand, reason about, and interact with the physical world.

GuardianAI is **not** intended to be another object detection demo.

Its purpose is to expose the entire perception pipeline so students can see how intelligence emerges from a sequence of understandable steps.

Why GuardianAI Exists

Modern AI tools are becoming increasingly capable, but they also risk becoming black boxes. Students can call an API or run a model without understanding how the system actually works.

GuardianAI was created to reverse that trend.

Instead of hiding complexity, it reveals it progressively.

The objective is to transform students from AI users into AI builders.

Relationship to ETHOS

ETHOS exists to help students become builders.

GuardianAI is the flagship project for introducing embedded AI because it naturally combines software, electronics, cameras, sensors, decision making, and hardware control into a single learning journey.

Relationship to PiPal-AI

PiPal-AI is envisioned as a collection of reusable AI capabilities for Raspberry Pi projects.

GuardianAI provides the Vision capability.

Future capabilities may include:

- GPS
- SDR
- Robotics
- Audio
- Environmental Sensors

All follow the same philosophy:

Sense $\rightarrow$ Understand $\rightarrow$ Reason $\rightarrow$ Act

Educational Philosophy

GuardianAI follows one central principle:

Reveal one new idea at a time.

Students should never feel that AI is magical.

Every lesson introduces exactly one new concept while reinforcing earlier concepts.

Layers of Perception

Layer	Name	Primary Question
0	Reality	What does the camera see?
1	Brightness	What happens when color disappears?
2	Simplification	What happens when detail disappears?
3	Boundaries	How are edges detected?
4	Change	What changed?
5	AI Vision	What image reaches the neural network?
6	AI Hypotheses	What predictions did the network make?
7	Understanding	Which objects are present?
8	Localization	Where are the objects?
9	Reasoning	What should the computer do?
10	Action	How does software affect the physical world?

Long-Term Vision

GuardianAI is intended to become the reusable Vision subsystem inside PiPal-AI.

The same architecture should support projects such as:

- Garden Guardian
- Parking Assistant
- Wildlife Monitor
- Face Recognition
- Rover Vision

Only the decision logic and hardware actions should change.

Success Criteria

A successful GuardianAI student should understand:

- How a camera captures images.
- How images become tensors.
- How neural networks recognize patterns.
- How predictions become objects.
- How software reasons about the environment.
- How decisions become physical actions.

The ultimate outcome is confidence.

Students should leave believing they can design and build intelligent machines of their own.

Guiding Principles

1. Education before optimization.
 2. Raspberry Pi first.
 3. Explain every stage.
 4. One responsibility per module.
 5. Reuse the Vision Pipeline.
 6. Keep perception, reasoning, and action separate.
 7. Build curiosity before complexity.
 8. Documentation evolves with the software.
-

Lessons Learned

GuardianAI is not about object detection.

Object detection is simply the first vehicle used to teach a much larger idea: how intelligent systems observe, interpret, decide, and act.

As the platform evolves, implementation details will change, but this educational philosophy should remain constant.